



Prediction of listening effort ratings for habitual and clear-Lombard speech presented in noise

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Abstract

Earlier research developed the high-energy glimpse proportion metric (HEGP) to capture those speech-dominant spectro-temporal regions, or glimpses, that survive energetic masking when speech is presented in noise. Whereas the HEGP metric has been shown to correlate with intelligibility of speech presented in noise, this study investigated whether HEGP predicts listening effort (LE) ratings, and if so, whether it does so differentially for habitual and clear-Lombard speech. Secondly, the study investigated whether acoustic measures of spectral balance, median F_0 and F_0 range, and articulation rate, explained additional variance in LE. Results showed that HEGP predicted LE ratings equally strongly across speaking styles. Beyond HEGP, wider F_0 range and lower articulation rate were also associated with decreased LE, particularly in the habitual speech condition. Thus, the HEGP release-from-masking metric, complemented by speaking-style aspects, can help explain LE ratings in noise.

Index Terms: listening effort, glimpsing, masking, noise, F_0 , speech rate

1. Introduction

Earlier research developed the articulation index [1] and the subsequent Speech Intelligibility Index (SII; [2]) to account for the finding that speech intelligibility in noise is proportional to the amount of audible speech information in a number of frequency bands. Following up on an earlier glimpsing model proposed to predict speech intelligibility in noise [3], Tang and Cooke [4] developed the *high-energy glimpse proportion* metric (HEGP) to capture those speech-dominant spectro-temporal regions, or glimpses, that survive energetic masking when speech is presented in noise. In line with findings from other intelligibility metrics [5, 6], Tang and Cooke [4] showed that selecting only *high-energy* glimpses led to equal or improved correlations between predicted and subjective intelligibility (i.e., percentage keywords correctly identified by listeners), relative to taking *all* glimpses into account.

Though the HEGP metric has been shown to correlate with intelligibility across speech types and signal-to-noise ratios [4], its connection to listening effort (LE) remains unexplored. Importantly, LE measures may capture differences in ‘ease of processing’ across various listening conditions, even when speech is maximally intelligible. These measures can thus help quantify differences in the effort required to understand what has been said. LE has been operationalized in different ways [7], including behavioral measures (e.g., response times; [8]),

physiological measures (such as pupillometry [9]), or self-report measures, such as questionnaires or rating scales (e.g., [10, 11]). A recent study by Simantiraki and colleagues [12] explicitly investigated the relationship between intelligibility, self-reported LE, and a pupil dilation measure for different speech conditions, including artificially-enhanced and synthetic speech conditions. Simantiraki et al. found that intelligibility scores were inversely related to subjective LE ratings, whereas the relationship between intelligibility and pupillometry data was more complex. Given this inverse relationship observed in [12], HEGP scores in the present study might be expected to predict subjective effort ratings.

When speaking in the presence of noise, talkers generally adjust their speech to (partly) overcome the noise. Talkers producing this so-called ‘Lombard’ speech [13-14] are known to increase vocal effort and slow down speech rate. Shen et al. [15] reported an acoustic analysis of 78 talkers producing speech in two speaking styles: a habitual style, and a style where talkers spoke in the presence of loud noise and were instructed to speak clearly. Given the combination of noise exposure and explicit clear-speech instruction, this style was termed ‘clear-Lombard style’ to differentiate it from simple Lombard or clear speech. As expected, speech in the clear-Lombard style was slower, exhibited a higher and more variable F_0 , and was produced with less spectral roll-off than habitual speech.

Following up on this acoustic analysis [15], the present study collected listening effort ratings for a subset of the same speech samples, presented to raters in the same speech-shaped noise that had been used to elicit the clear-Lombard speech in [15]. Specifically, we investigated whether human LE ratings can be predicted from HEGP scores for the two speaking styles (habitual and clear-Lombard style; thereby including conditions where talkers do and do not adapt their speech to the background noise), and if so, whether HEGP’s predictive power differs for the two styles (Research Question 1).

However, note that HEGP only quantifies the contribution of energetic masking to intelligibility or LE, while other acoustic measures may reflect different aspects of between-talker and between-speaking style differences related to speech clarity and, hence, to ease of processing. For instance, [15] focused on four acoustic-phonetic measures: articulation rate, median F_0 , F_0 range, and spectral balance differences between the two speaking styles. These acoustic-phonetic measures may explain additional variance in LE ratings, not accounted for by HEGP, and some of them have been related previously to intelligibility gains in Lombard or clear speech (cf. [16], for a review of context-induced speech modifications and their impact on intelligibility). Therefore, our second research

question (RQ2) asks, *if* HEGP predicts listening effort ratings, do sentence-level acoustic-phonetic measures (i.e., articulation rate, F_0 median and F_0 range, and spectral balance) explain *additional* variance in listening effort ratings? In line with RQ1, RQ2 also asks whether the strength of the relationship between LE ratings on the one hand and acoustic measures on the other differs between habitual and clear-Lombard styles.

Given the overlap between the release-from-masking nature of the HEGP metric and spectral balance, the latter is not expected to explain much additional variance in listening effort. Likewise, mean F_0 was shown not to relate to sentence intelligibility if gender-dependent intelligibility differences are accounted for [17], thus it is not expected to explain additional variance in listening effort ratings. Slower articulation rate [18–20, but cf. 17], and enlarged pitch range [17, but cf. 21], on the other hand, may help listeners to understand speech in noise with less effort, possibly because they serve as proxies for a clearer speaking style [22]. We addressed our research questions by presenting a subsample of the materials from [15] in an online rating study.

2. Methods

2.1. Participants

Listening effort ratings were collected from 230 young adult normal-hearing participants via a university participant portal. Participants were all Radboud University students, aged between 18 and 30 years, normal-hearing and native speakers of Dutch (as per self-report). Participants were reimbursed for their participation with a digital gift voucher after a response validity check (a check that participants did not provide the same rating throughout the experiment). Ethics consent was obtained for the rating procedure from the Ethics Committee for the Humanities at Radboud University. Informed consent was obtained from all raters prior to their involvement in the study.

2.2. Speech materials

The speech materials were a subset of the multi-talker sample reported in [15], in which talkers read the same sentence set twice: first in their habitual reading style, and then in loud speech-shaped noise presented over headphones. Talkers were instructed to speak as clearly as possible when reading in noise (i.e., clear-Lombard style). Talkers read the same sentences from the sentence set twice in two different random orders: one for the habitual and one for the clear-Lombard style. All 48 test sentences were between 12 and 16 syllables long and were syntactically correct and semantically plausible. An example sentence is ‘Ferdie heeft het boek helaas in het park laten liggen’ (‘Ferdie has unfortunately left the book in the park’).

For the rating of listening effort, we wanted to avoid presenting the same sentence content to raters more than once, as listening effort may be lower upon hearing the same content repeatedly. As our materials involved two renditions (reading styles) of the 48 unique sentences from 78 talkers, each rater could only rate 48 utterances (sampled from different talkers and the two styles). Out of the total sample of 78 talkers, speech materials of a subset of 48 talkers were selected in a pseudorandom fashion. That is, the only selection restriction was that twice as many female (32) as male talkers (16) were selected, in line with the unequal gender distribution in the total sample. The total number of utterances presented in the online rating study thus amounted to 48 (talkers) * 48 (unique sentences) * 2 (speaking conditions) = 4608 utterances.

Each participant in the current study rated 48 unique utterances from six different talkers. Specifically, each rater was presented with eight utterances per talker: four in the habitual speaking style and four in the clear-Lombard style. Utterances were distributed over 96 different experimental lists to accommodate multiple combinations of talkers, speaking styles, and sentence content.

All 4608 utterances were scaled to an average intensity of 65 dB SPL and mixed with speech-shaped noise of 71 dB SPL into a mono recording, resulting in an SNR of -6 dB. The noise started 500 ms before sentence onset and lasted till 500 ms after sentence offset. Additionally, there were 200 ms fade-in and fade-out times only affecting the noise in the recording.

Following [4], HEGP scores were calculated for all 4608 utterances for presentation in speech-shaped noise at -6 dB SNR. Note that the speech-shaped noise for HEGP calculation and speech presentation was the same noise used to elicit the clear-Lombard speaking style during reading aloud [15]. HEGP scores range between 0 and 1, with higher numbers indicating a higher glimpse proportion escaping energetic masking, in turn indexing higher predicted intelligibility.

Sentence-level acoustic measures (articulation rate in syllables per second, F_0 median and range, and spectral balance) were available for these materials from [15]. Spectral balance was calculated by extracting the long-term average spectrum of each sentence with a filter bandwidth of 100 Hz and subtracting the energy maximum in the low-frequency range (0–2 kHz) from that in the high-frequency range (2–5 kHz), with higher values indicating less vocal effort. (Median) F_0 values were converted to semitones, with 1 Hz as reference value. For F_0 range, the difference between the 10th and 90th percentile was computed for each sentence.

2.3. Procedure

Upon receiving and clicking the link to the online study, participants provided informed consent for their anonymous rating data to be used for research purposes. Participants were only directed to the rating study if they met the inclusion criteria concerning age, hearing and native speaker status (cf. 2.1).

Participants were randomly assigned to one of 96 experimental lists. They were presented with four practice items (identical across lists) to familiarize themselves with the task procedure and noise level. Practice items were taken from four different talkers (3 female and 1 male), who formed a subset of the test-material talker set. Practice items differed in lexical content from the 48 test sentences but were mixed with speech-shaped noise at the same SNR as the test materials. Participants were instructed to listen to the materials through headphones and to adjust the presentation level to an optimal level for sentence understanding. Participants could only listen to each utterance once before providing their rating (during practice and test) and could only proceed to the next trial after clicking the play button and providing their listening effort rating.

Participants rated their listening effort to understand the sentence content of the utterance on a scale from 1 to 7 with 1 representing ‘not effortful at all to understand’ and 7 representing ‘extremely effortful to understand’. This scale was an adapted version of the *Adaptive Categorical Listening Effort Scaling Test* developed by Krueger et al. [10]. The phrasing of listening effort here thus also includes understanding the words themselves, and hence the notion of comprehensibility. Note that Krueger et al. [10] included the option for raters to respond with ‘only noise’ when no speech cues were perceived,

complementary to the 7-point effort scale. Given our set-up with a fixed signal-to-noise level, which was based on pilot testing to yield, on average, medium effort levels across styles, this ‘only noise’ option was not included in our set-up. The verbal labels representing the scale extremes were visible throughout the test. Participants clicked one of seven radio buttons to provide their rating. Participants were instructed to consider whether they could understand part of the sentence and to provide a rating of 7 if they had hardly understood anything, and to choose a lower rating if they had understood at least some words. The presentation order of utterances on a specific list was randomized for each participant.

2.4. Statistical analysis

The obtained ratings of the 230 raters were analyzed with linear mixed-effects models in *RStudio* (R version 4.3.1) with the *lme4* package (version 1.1-15). To address the first research question (RQ1), we fitted models without additional acoustic measures to only investigate how listening effort rating was predicted by HEGP. Additionally, RQ1 addressed whether this relationship between listening effort rating and HEGP was modulated by speaking style, thus testing for the effects of HEGP score and Speaking style as the fixed effects of interest, and their interaction, on listening effort ratings. Talker gender was added as a control variable. Additionally, we compared a model with Talker gender as a main effect to a model in which Talker gender interacted with Speaking style to test whether male or female talkers made larger clear-Lombard adjustments to clarify their speech. Models were compared with the *anova()* function in R to decide on the best fitting model. Across models, Sentence ID, talker ID, and rater ID were included as random effects, and we also included a by-talker random slope for Speaking style to account for the fact that talkers differed in the extent to which their two speaking styles differed. Sum coding was used for contrast coding of Speaking style and Gender, and all continuous predictor variables were mean-centered.

As a second step, to answer the second research question (RQ2), we took the best fitting model from step 1 and compared its fit to that of a model in which four speech acoustics measures (articulation rate, median F_0 , F_0 range, and spectral balance) were added as continuous (mean-centered) predictors to see whether their inclusion could explain additional variance in listening effort ratings. Specifically, we tested whether the speech measures were differentially predictive of listening effort, depending on speaking style (thus testing for their interaction). Lastly, if their (joint) inclusion improved model fit, we also tested *which* speech measure(s) improved model fit.

3. Results

The 230 raters each rated 48 utterances, which amounted to 11040 ratings. Consequently, each of the 4608 utterances, which represented a combination of sentence, talker and speaking style, was rated by two or three raters. Descriptive data on listening effort ratings, HEGP predicted intelligibility scores, and acoustic measures for the habitual and clear-Lombard speaking style materials are provided in Figure 1. Figure 1 provides a histogram for the discrete rating data, and density plots for all continuous (speech-material) measures. Note that the bimodal distribution of F_0 median reflects the inclusion of male and female talkers. Compared to the habitual speaking style, effort ratings are lower and HEGP is higher for the clear-Lombard style. As mentioned earlier, speech in the clear-Lombard style was slower, exhibited a higher and more

variable F_0 , and was produced with less spectral roll-off than habitual speech [15]. Correlation strength (Pearson r) between HEGP and each of the four acoustic measures ranged between 0.35 (F_0 range) and -0.81 (spectral balance), across styles.

A comparison of the models that were run to address RQ1 showed that Talker gender did not interact with Speaking style. The simpler model was therefore selected, in which Gender was added as a control variable, without its interaction with Speaking Style. Results from this simpler Step 1 model are illustrated in Figure 2, and show that listening effort ratings can be predicted from HEGP ($\beta = -8.93$ (SE = 0.41), $t = -21.93$, $p < 0.001$), where higher HEGP scores relate to lower perceived listening effort, and from Speaking Style, with clear-Lombard style leading to lower perceived listening effort than habitual style ($\beta = -0.55$ (SE = 0.08), $t = -7.14$, $p < 0.001$). However, the predictive power of the HEGP metric did not differ across speaking styles ($\beta = 0.58$ (SE = 0.61), $t < 1$, *n.s.*). The Gender effect ($\beta = 0.81$ (SE = 0.11), $t = 7.15$, $p < 0.001$) showed that male voices were rated more effortful to listen to than female voices, and this held equally across styles.

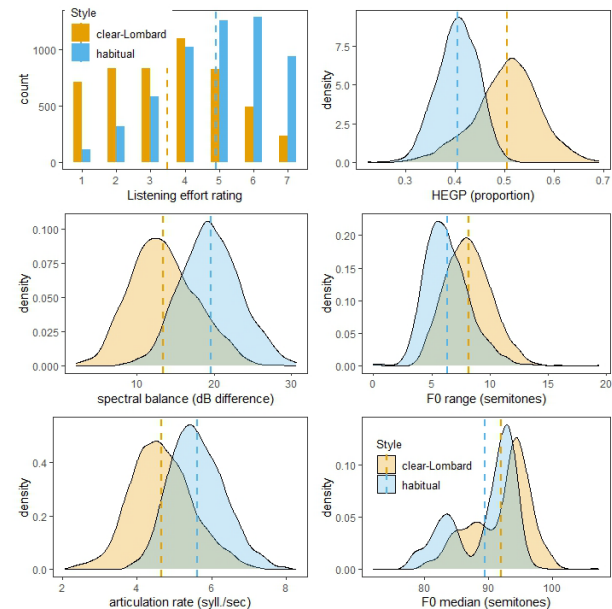


Figure 1: Distribution data for listening effort ratings ($N = 11040$, top left panel) and speech-material measures (for each of which $N = 4608$), split by speaking style. Dotted lines represent the distribution means per speaking style.

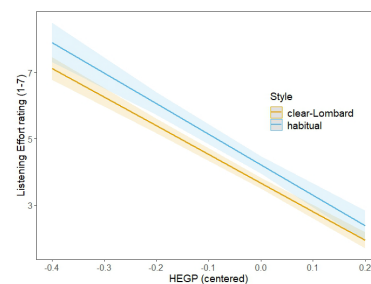


Figure 2: Step 1 model plot showing the association between listening effort rating and HEGP for speech (presented in noise), across the two speaking styles.

In order to address RQ2, we took the Step 1 model, and added the four speech (acoustic-phonetic) measures, and their interaction with speaking style. The resulting Step 2 model had a significantly better fit than the Step 1 model (cf. section 2.4). The Step 2 model is presented in Table 1. Three out of the four additional acoustic measures surfaced as significant predictors of listening effort. That is, an increased pitch range decreased listening effort, and higher values for spectral balance (indicating steeper spectral roll-off), as well as higher articulation rates, were associated with increased listening effort. Like HEGP, spectral balance was equally predictive of listening effort, regardless of speaking style. However, the relationships between F_0 range and articulation rate on the one hand, and listening effort on the other, were modulated by speaking style: both acoustic measures were more strongly associated with listening effort for habitual speech.

Table 1: *Step 2 model predicting listening effort ratings. Significance level notation indicates that $p < 0.05$ (*); $p < 0.01$ (**); or $p < 0.001$ (***)*.

Step 2 model parameters		
<i>Fixed effects</i>	<i>Estimate (SE)</i>	<i>t (sign. level)</i>
HEGP	-8.42 (0.50)	-16.71 (***)
(Speaking) Style	-0.39 (0.082)	-4.67 (***)
HEGP x Style	-0.068 (0.80)	-0.08
Gender	0.80 (0.13)	6.21 (***)
F_0 range	-0.036 (0.008)	-4.56 (***)
Spectral balance	0.011 (0.0052)	2.10 (*)
Articulation rate	0.094 (0.033)	2.82 (**)
F_0 median	0.0037 (0.0077)	0.48
F_0 range x Style	0.036 (0.015)	2.45 (*)
Spectral bal. x Style	-0.00015 (0.0095)	-0.02
Artic. rate x Style	-0.19 (0.047)	-4.10 (***)
F_0 median x Style	0.00068 (0.011)	0.06
<i>Random effects</i>	<i>Variance (SD)</i>	
Rater	0.44 (0.66)	
Talker	0.12 (0.35)	
Style x Talker	0.17 (0.41)	
Sentence	0.07 (0.26)	

As a last step, we analyzed whether each of the three (additional) acoustic measures that surfaced as significant predictors of listening effort in the Step 2 model individually improved model fit, relative to the Step 1 model. This was indeed the case for articulation rate and F_0 range, but not for spectral balance, in line with our expectation. Hence, articulation rate and F_0 range, if added as main effects, but even more so if allowed to interact with speaking style (cf. Table 1), explained variance in listening effort, beyond the predictive power of energetic masking level (HEGP).

4. Discussion

This study investigated whether listening effort ratings, for speech presented in speech-shaped noise, could be predicted by an energetic masking measure that focuses on high-energy glimpses (HEGP, [4]). In line with our expectation, an inverse relationship between HEGP and subjective listening effort was observed. Moreover, this association between HEGP and effort held equally strongly for speech produced habitually and for speech produced in a clear-Lombard style. Secondly, as HEGP only reflects (escape from) energetic masking, we asked

whether acoustic measures related to speech clarity, rather than masking, would explain additional variance in LE ratings for both speaking styles. Indeed, adding articulation rate and pitch range significantly improved model fit, and hence accounted for additional variance in effort ratings.

These latter results are in line with findings on associations between speech rate (e.g., [19]) and F_0 range on the one hand and intelligibility on the other [17]. That is to say, slower rate and extended pitch range may not only be the best proxies for careful articulation (out of the included measures), extended pitch range may also help listeners to keep their attention focused on the target speech. Additionally, our observation that median F_0 did not predict listening effort if gender-dependent intelligibility differences are accounted for is in line with previous literature linking acoustic variation to speech intelligibility in noise [17].

The observation that male talkers were rated as more effortful to listen to than female talkers could relate to actual ‘inherent’ differences in how clearly (the selected) male and female talkers speak, or to different amounts of energetic masking. Specifically, even though the speech-shaped noise was based on equal numbers of male and female talkers, the noise’s spectral composition may have masked relatively more of the speech content of the male than female talkers. A post-hoc t-test showed evidence that averaged HEGP scores for the female-talker materials (0.47) were indeed higher than those for male talkers (0.42), explaining at least part of the gender effect.

Note that more variation was observed in clear-Lombard than habitual speech, in terms of listening effort values, and in HEGP values, as well as articulation rate and F_0 range values, as predictors of listening effort (see wider distributions for clear-Lombard style in Figure 1). In principle, this greater variation could allow associations to show up more strongly for clear-Lombard speech. Such a result would also be in line with results by Paulus and colleagues [23], showing talker-acoustics effects on listening effort quantified by pupillometry for more intelligible speech conditions, but not for less intelligible speech conditions (with less variation in pupil dilation). However, whereas the predictive strength of HEGP for listening effort remained the same across speaking styles, articulation rate and F_0 range were found to be more strongly associated with listening effort for habitual-style speech than for clear-Lombard speech. One interpretation of these results is that HEGP (as well as spectral balance, cf. Table 1), representing speech audibility, can be considered an important first prerequisite for linguistic processing, and therefore equally important across speaking styles. Given sufficient speech audibility, speech clarity differences (between talkers and between sentences) may then matter more for habitual than clear-Lombard speech, possibly because of the overall lower clarity of habitual speech.

5. Conclusion

In the current study native listeners provided listening effort ratings for two forms of speech, habitual and clear-Lombard, presented in speech-shaped noise. The principal novel finding was that an existing objective intelligibility metric (HEGP), complemented by speech-clarity related acoustic measures of F_0 range and articulation rate, can help explain listening effort ratings in noise across speaking styles. These findings also provide further evidence for the link between objective intelligibility and subjective ratings of listening effort.

6. References

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